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21 June 1983
SUPERSEDING
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6 December 1972
(See 6.6)

MILITARY SPECIFICATION

DEGREASER, VAPOR SPRAY, STATIONARY, SOLVENT-TYPE, STEAM OR ELECTRICALLY-HEATED

This specification is approved for use by the US Army Armament Materiel Readiness Command, Department of the Army, and is available for use by all Departments and Agencies of The Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hand-operated, electrically or steam-heated degreasers for use with hydrocarbon type solvents for the removal of dirt, grease, oil, and sludge from metal and other surfaces.

1.2 Classification. Degreasers shall be of the following classes and sizes as specified (see 6.2).

Class 1-- Electrically heated.

Size 1-- Minimum inside dimensions: length, 70 inches; width, 30 inches; with 30-inch vapor depth.

Size 2-- Minimum inside dimensions: length, 84 inches; width, 36 inches; with 36-inch vapor depth.

Class 2-- Steam heated.

Minimum inside dimensions: length, 70 inches; width, 30 inches; with 30-inch vapor depth.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Materiel Readiness Command, ATTN: DRSAR-LEE-S, Rock Island, IL 61299 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4940

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2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-A-548	-	Antifreeze/Coolant, Engine, Ethylene Glycol, Inhibited, Concentrated
CC-M-1807	-	Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller)
QQ-S-781	-	Strapping, Steel, and Seals
TT-P-636	-	Primer Coating, Alkyd, Wood and Ferrous Metal
PPP-B-636	-	Box, Shipping, Fiberboard
PPP-C-843	-	Cushioning Material, Cellulosic
PPP-T-60	-	Tape, Packaging, Waterproof

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-T-152	-	Treatment, Moisture and Fungus Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-P-775	-	Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles, and Strainers
MIL-E-16298	-	Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of
MIL-I-24092	-	Insulating Varnish, Electrical, Impregnating, Solvent Containing
MIL-C-52950	-	Crate, Wood, Open and Covered

STANDARDS

FEDERAL

FED-STD-H28	-	Screw Thread Standards For Federal Services

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-889	-	Dissimilar Metals
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods

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* 2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CRF 1910).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA Electrical Standards

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, New York 10017.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one complete degreaser for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 3.2 Design. The degreaser shall be new and of the latest standard commercial design. The degreasers shall be designed for use with the chlorinated hydrocarbon type of solvent. The solvent is heated to its boiling point

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producing a vapor which fills the tank to the level of the condenser. The condensate is continuously collected and conducted from the condenser to a water separator from which water-free solvent is piped to the solvent storage tank. Overflow from the solvent storage tank is returned to the boiling chamber, thus completing the distillation cycle. A motor-driven pump shall be provided for spraying clean solvent on work for additional cleaning and cooling purposes. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement.

3.2.1 Maintainability. Design and configuration of the degreasers shall insure that maintenance tasks required for sustaining normal performance and operation shall be reduced to the simplest form and procedure. Common tools and equipment shall be used. Instructions for disassembly, repair, reassembly, and adjustment shall be clear, concise, and definitive.

* 3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.4 Construction. The degreasers shall be complete and so constructed that when installed, they can be used for the operations specified herein.

3.4.1 Castings and forgings. All castings shall be free of blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclosures, cold shuts, mismatching, and other defects that could affect the structural strength of the forgings involved. Strength and other essential physical and chemical properties of the castings and forgings shall meet the performance requirements specified herein.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be performed in accordance with standard commercial practice. In no event shall such processes, including peening or plugging, be used on castings or forgings for reclaiming any defective parts of the degreasers.

3.4.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. All such parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.4.4 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the connections shall be mechanically secured before soldering. Rosin base fluxes only shall be used in soldering operations. Connections to screw-type terminals shall be mechanically secured to prevent loss of tightness.

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* 3.4.5 Threads. Threaded parts shall conform to Federal Standard H28 and the applicable Detailed Standard section referenced therein.

* 3.4.6 Gears. All gears and pinions shall be constructed from material suitable for their intended purpose. The gears and pinions shall be machined and heat treated by a process that shall impart the hardness, smoothness, and toughness that shall enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.4.7 Lubrication. All bearings (except sealed-for-life type) and other working parts shall be provided with standard means for lubrication.

* 3.4.8 Safety and health requirements. Safety devices shall be provided for all parts of the degreaser presenting safety hazards. Guards shall be removable to provide easy access to the guarded parts. The degreaser shall comply with the safety and health requirements promulgated under OSHA 2206, (Title 29, Code of Federal Regulations, Chapter XVII, part 1910) and amendments, that are applicable to degreaser itself.

3.4.9 Interchangeability. All parts subject to replacement shall be designed and manufactured to such standards as shall permit replacement without the need for modification of the parts or the degreaser.

* 3.4.10 Surfaces. Surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be clean and free of sand, dirt, fins, sprues, scale, flux, and other extraneous materials. External surfaces shall be smooth and all edges shall be rounded or beveled except as otherwise required by design.

3.5 Performance and product characteristics.

3.5.1 Classes and sizes. Class 1 and 2 degreasers shall conform to the following:

3.5.1.1 Class 1, size 1. Degreasers of this class and size shall be electrically heated and provided with equipment for spraying clean solvent. Continuous rate of vaporization shall be produced by thermostatically controlled electric heating elements. Solvent capacity of the size 1 degreasers shall be at least 90 gallons. The no-load distillation rate shall be at least 30 gallons per hour.

3.5.1.2 Class 1, size 2. Degreasers of this class and size shall have a solvent capacity of at least 105 gallons, including the boiling tank and spray pump reserve combined. The no-load distillation rate shall be at least 30 gallons per hour.

3.5.1.3 Class 2. Degreasers of this class shall comply with all requirements of the class 1, size 1 degreasers except for the following: saturated steam at the proper pressure and temperature, through immersed closed coils, shall be used to produce continuous vaporization of the solvent at all increments of demand. A steam valve shall be provided a with thermostat control for automatic shut-off should the vapor level for any reason rise above the level recommended by the manufacturer.

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3.5.2 Heating capacity, Class 1 and Class 2. The degreasers of both of these classes shall be capable of raising the temperature of the solvent from 60° Fahrenheit (F) to its boiling point in not more than one hour, with the boiling chamber loaded to rated solvent capacity. The degreasers shall have a heat recovery sufficient to degrease at least 3000 pounds of evenly distributed metal parts covered with heavy deposits of sludge, gum, or grease per hour.

3.6 Details of components.

3.6.1 Cleaning tank. The cleaning tank for the degreasers shall be fabricated from stainless steel or clad sheet stainless steel, at least Manufacturer's Standard Gage No. 10. The tank shall be braced or stayed where necessary to prevent buckling or deformation of the sides. A flanged cleanout opening shall be provided at one end of the tank of such size and location as to permit removal of sludge, grease, and other semi-solid deposits from the tank interior. The cleanout shall be large enough to allow insertion and removal of the electric heaters or steam coils into and out of the tank. The cleaning tank and all components employed in the storing or handling of water or solvent shall be leakproof. A drain cock or valve shall be provided near the cleanout opening for completely draining the liquid contents of the tank. This cock or valve shall be removable, shall not be permanently attached to the tank, and shall be of such size as to drain the tank in not more than ten minutes. The tank bottom shall be constructed to slope toward the tank drain valve and cleanout opening. Provisions shall be made in the tank for attaching thermostats and capillary bulbs and protecting them from possible damage. The cleaning tank shall be provided with a metal cover to completely cover the top of tank when not in use.

3.6.2 Work supports. Removable work supports shall be furnished of such size as to cover the entire tank bottom and located to provide protection to the heating units and prevent immersion of parts in dirty solvent. The work supports shall consist of at least two sections and not more than four, any or all sections of which may be removed. Construction shall be of stainless steel, expanded metal, or open bar grating. The work supports shall be supported to prevent sagging or buckling when an area of four square feet, located in the center of the support, is subjected to a unit loading of 250 pounds per square foot.

* 3.6.2.1 Work baskets. Each degreaser shall be furnished with two stainless steel work baskets of expanded metal type construction of sufficient strength to support at least 100 pounds per square foot. The baskets furnished shall be one full-size and one half-size basket. The baskets shall be provided with a handle(s) for hoisting. The full size basket shall have length, width and height two inches smaller, ±1/2 inch, than the nominal inside dimensions of 1.2. The half size basket shall have a length 50 percent of the nominal inside length and width and height as specified above (see 1.2).

3.6.3 Cleanout door. A cleanout door of the proper size to fit the flanged cleanout opening in the tank shall be provided. The door shall be fabricated from stainless steel sufficiently heavy to prevent warping or bending and strong enough to rigidly support either steam coils or the electric heaters. A gasket fabricated from solvent and heat-resistant material shall be furnished. Sufficient screws or bolts shall be used in securing the door to the tank to insure against solvent leakage.

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3.6.4 Condenser. The condenser shall be of a capacity to control the solvent vapor at all times with an inlet water temperature of 75°F or less and an outlet temperature of from 90°F to 120°F. The condenser assembly shall consist of a water jacket and series of coils. The water jacket shall completely encircle the cleaning tank and shall be located to provide the specified vapor depth. The coils and pipe fittings shall be fabricated from stainless steel. The installation of coils shall be such that they may be removed and replaced. Water jackets shall be securely attached to the cleaning tank. All seams shall be welded with stainless steel rod or wire. The condenser assembly (water jacket and coils) shall be capable of withstanding a hydrostatic pressure of at least 40 pound force per square inch gage (psig) for not less than 30 minutes. A stainless steel valve shall be furnished for the condenser water shutoff. The valve shall be removable for replacement or repair. Inlets and outlets shall be arranged so that the condenser assembly is completely filled with water at all times during the operation of the degreaser. The water jacket shall be provided with removable plugs at each corner, near the bottom, to facilitate cleaning and removal of lime or other deposits from the inside surfaces. All condenser water outlets shall be piped to a common drain emptying into an open sight flow device. The condenser system shall be electrically interlocked with the heater so that the heater cannot be turned on without water flowing through the condenser.

3.6.5 Open sight flow device. The condenser outlet shall be provided with an open sight flow device to allow the operator to visually determine that water is flowing through the condenser. The device shall be constructed from stainless steel.

3.6.6 Electric heaters. Class 1 degreasers shall be furnished with immersion-type electric heaters of sufficient heating capacity to permit continuous operation of the machines at the cleaning rate specified in 3.5.2. The immersion-type heaters shall be attached to the cleanout door or may be of the portable type that shall be easy to insert and remove. The heaters shall be furnished with sealed, stainless steel casings and shall be protected from burnout when little or no circulation of the heated liquid occurs around the heater or when solvent level is low. The electric terminals shall be equipped with screws and clamps of a size for connection to the power supply and shall be provided with a cover to prevent possible contact by operating personnel or damage to the connections. The heaters shall be located to be completely covered with solvent under normal operating conditions. The heaters shall operate from 230 volts, 60 hertz (Hz), three phase.

* 3.6.7 Steam coils. Class 2 degreasers shall be furnished with steam coils of sufficient heating capacity to permit continuous operation at the capacities specified in 3.5.1.3. The coils shall be welded to the cleanout door and shall have the inlet and outlet threaded for connection to the steam supply and return lines. The coils shall be fabricated from stainless steel. Coils shall be located so that they are completely immersed in solvent at all times under normal operating conditions. The coils shall be capable of withstanding a hydrostatic test pressure of at least 75 psig for a minimum of 30 minutes.

3.6.8 Condensate trough. A condensate trough or pan for the collection and return of condensate to the water separator shall be provided with each degreaser. The trough or pan shall be capable of accommodating the highest rate of condensation for the degreaser and shall be located beneath the coil. It shall

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be fabricated from the same corrosion-resistant material and gage as the tank. An outlet to the water separator shall be provided in the wall of the cleaning tank.

3.6.9 Precooler. A water-jacketed precooler shall be installed in the water separator system. The precooler and water separator may be an integral unit. The precooler shall be capable of preventing the vapor condensate from entering the separator until the temperature has dropped to 160°F or lower.

* 3.6.10 Water separator. A water separator of the gravity type shall be furnished with each degreaser. The separator shall be mounted on the outside wall of the degreaser, accessible for periodic maintenance. It shall be of such capacity as to efficiently separate the solvent from the condensate returns during peak rates of condensation for the particular degreaser on which it is installed. Construction shall be of stainless steel. A drain plug and vent opening shall be provided when the design is such as to require it. The direction of flow through the separator shall be permanently indicated on the body of the separator.

3.6.11 Automatic condenser water temperature control valve and strainer. All degreasers shall be equipped with an automatic condenser water temperature control valve of the temperature or pressure-actuated type. The valve shall be stainless steel and capable of handling the maximum cooling water requirements of the degreaser. The valve shall control the flow of cold water into the condenser and shall be operated by the temperature of the water at the condenser outlet. This valve shall be capable of operating over a minimum temperature range of from 80°F to 120°F. Means shall be provided for manual adjustment of the maximum flow of cold water through the valve. An adjustable or fixed type of bypass shall also be incorporated in the valve to permit a small amount of water to pass the thermal or pressure control element of the valve at all times while the degreaser is in operation. A sediment strainer shall be installed in the cold water inlet side of the temperature control valve. The strainer screen shall be fabricated from stainless steel. The direction of flow shall be permanently indicated on the bodies of both the temperature control valve and the strainer. The temperature control valve shall be capable of controlling the water temperature to within ±5°F of the preset condenser outlet temperature.

3.6.12 Dial-indicating temperature gage. All degreasers shall be equipped with a dial-indicating temperature gage. The dial shall not be more than six inches in diameter with numerals and graduations clearly marked. The temperature range of the instrument shall cover a minimum range of from 100°F to 300°F and be numbered every 20°F or 25°F, with scale divisions every 2°F or 5°F, respectively. The case shall be of stainless steel and of either the wall or flush-mounting type. The indicator gage shall be mounted on the degreaser where it is visible to the operator at all times and not in direct contact with the heated surface of the cleaning tank. The capillary tube shall be of a type suitable for mercury-filled instruments and shall be properly armored and attached to the exterior of the cleaning tank. The bulb shall be located in the boiling solvent area of the tank. The bulb shall be provided with a protective guard. Means shall be provided for calibrating or zeroing the gage when necessary. The accuracy of the instrument shall be within ±5°F.

3.6.13 Solvent level gages. Solvent level gages shall be furnished to indicate the level of the boiling solvent in the cleaning tank and the amount of solvent in the storage tank. The gages shall be of stainless steel, adequate to

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cover the maximum and minimum solvent levels encountered at those locations during normal operation of the degreaser. Stainless steel shut-off valves shall be provided at the top and bottom of the gage. A stainless steel drain cock shall be furnished at the bottom. The glass components shall be provided with guards. The level gages shall be capable of withstanding the maximum temperature to be encountered during normal operations.

3.6.14 Storage tank. The storage tank shall be of welded construction, using the same material as the cleaning tank (see 3.6.1). All seams shall be welded with stainless steel rod or wire. The storage tank capacity shall be equal to or greater than the solvent capacity of the machine. The tank shall be braced or stayed where necessary to prevent buckling or deformation of the sides. The storage tank may be attached to the cleaning tank or it may be an integral part of the cleaning tank. A stainless steel valve shall be located in the line from the storage tank to the boiling chamber for use in distillation and purification of the solvent. This valve shall be removable for replacement or repair.

3.6.15 Spray hose assembly. Degreasers shall be furnished with a spray hose assembly consisting of a flexible, solvent-resistant hose, a spray pipe or lance and nozzle, a hand operated shut-off valve located in or near the handle, and a durable solvent and heat-resistant handle. The length of the hose assembly shall be such that any part of the cleaning tank interior may be reached with the nozzle. The size of the hose and spray pipe shall be adequate for the pump capacity and nozzle design. The hose shall be furnished with adapters for connection to the pump outlet line and the spray pipe or lance.

3.6.16 Spray pump. The spray pump shall have a minimum capacity of at least four gallons per minute at an eight-foot head. The pump shall be located so that prime is maintained at all times. The pump shall be provided with anti-friction bearings. Packings and seals shall be of solvent-resistant material with means provided to compensate or replace for normal wear. The pump shaft shall be stainless steel and the impeller shall be either stainless steel or bronze. The pump motor shall conform to the requirements of 3.6.20.1.

3.6.17 Vapor level and solvent heat control.

3.6.17.1 Class 1. Class 1 degreasers shall be equipped with heat controls which are automatic and accurate in operation. The control system shall be provided with thermal overload protection with an automatic time delay relay or so arranged as to require manual resetting in order to continue operation after the temperature controller has broken the circuit. The system shall consist of the following: remote bulb temperature controller of the nonindicating type with two-position action employing a vapor tension thermal system with a stainless steel bulb and a metal armored capillary tube of proper length; on-off control switch; and a magnetic across-the-line contractor with overload protection, reduced voltage (115 volt) control circuit, and manual reset or automatic time delay relay.

3.6.17.2 Class 2. On class 2 steam-heated degreasers, the temperature controller shall actuate the hydramotor or solenoid-type valve, requiring at most 45 seconds to close. The temperature controller shall be provided with means for temperature adjustment. The temperature range of the controller shall cover a minimum temperature range of from 150°F to 300°F. Accuracy of control shall be

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within ± five percent of the temperature range. The capillary bulb shall be provided with a protective screen or guard or placed in a location out of the working area of the unit.

* 3.6.18 Freeboard height. Unless otherwise specified (see 6.2), degreasers shall have a freeboard height above the condenser of at least 75 percent of the inside, or working width, of the tank.

3.6.19 Warning plate. All degreasers shall be furnished with a warning plate of corrosion-resistant material permanently attached in a conspicuous location and at least 8 by 10 inches in size. Lettering shall be clear and legible and as large as possible for the plate size specified above. Each plate shall bear the following information:

WARNING

Maintenance of the tank shall not be undertaken unless all cleanout doors have been removed and all solvent and solvent vapors expelled, and only when equipped with protective equipment and observed by an assistant. Read operating instructions.

3.6.20 Electrical equipment.

* 3.6.20.1 Motor. The pump motor shall conform to the requirements of CC-M-1807, with class B insulation and continuous duty rating. The motor shall be induction squirrel-cage type and operate from a power source of 115 volts, 60 Hz, single phase.

3.6.20.2 Electrical components. The degreaser shall be completely wired to operate from a 230 volt, 60 Hz, three phase, four wire power source and be internally wired for operation of the 115 volt, 60 Hz, single phase motor and electrical equipment. All electrical components of the degreasers, including starters, relays, switches, wiring, and their inclosures, shall conform to, and shall be located in accordance with, the NEMA Electrical Standards except as specified herein. No component part shall be used in any application where the requirements exceed the manufacturer's rating for the part.

3.6.20.3 Fungus control. When specified (see 6.2), electrical components and circuit elements shall be treated for fungus resistance in accordance with type II requirements of MIL-T-152 except that the statement of treatment is not required. The windings of the electric motor shall be given two impregnating coats of varnish conforming to the requirements of MIL-I-24092, type M, class 130 through class 220 (manufacturer's option).

3.7 Painting. Unless otherwise specified (see 6.2), painting and finishing of the degreaser shall be in accordance with standard commercial practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall be clean and free of all foreign matter detrimental to painting and shall have at least one coat of primer and one coat of enamel, or equivalent, applied.

3.8 Identification marking. Degreasers shall be permanently marked for identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2), shall include the National stock number.

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3.9 Workmanship. Standards of workmanship shall assure that the degreasers shall have the stability and efficient operating characteristics found in standard commercial units and as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

* 4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

* 4.4 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Tests (see 4.6).
(c) Packaging inspection (see 4.7).

* 4.4.1 Sampling. When required (see 6.2), sampling for quality conformance inspection shall be as specified in MIL-STD-105 at the Inspection Level and Acceptable Quality Level (AQL) specified in the contract. For packaging inspection (see 4.7), special inspection level S-4 and an AQL of 4.0 percent defective shall be used.

* 4.5 Product examination. Each complete degreaser shall be examined to determine compliance with the requirements of Section 3. As a minimum, the following characteristics shall be examined:

CHARACTERISTICS	REQUIREMENT PARAGRAPH
Dimensions not as specified	1.2
Defective casting or forging	3.4.1
Insecure welding, brazing, or soldering	3.4.2
Insecure electrical connections	3.4.4
Defective threads	3.4.5

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Insufficient safety devices	3.4.8
Omission of any components	3.6
Cleaning tank fabrication not as specified	3.6.1
Work supports not removable or fabricated as specified	3.6.2
Work basket construction not as specified	3.6.2.1
Cleanout door not as specified	3.6.3
Electric heaters not as specified	3.6.6
Spray hose assembly not as specified	3.6.15
Spray pump not as specified	3.6.16
Warning plate omitted or not as specified	3.6.19
Electrical equipment not as specified	3.6.20
Painting not as specified	3.7
Identification marking not as specified	3.8
Workmanship defects	3.9

4.6 Tests.

4.6.1 Heating capacity test. The degreaser shall be loaded to its rated solvent capacity with the manufacturer's recommended cleaning solvent and the heater switch turned on. The solvent shall reach its boiling point in not more than one hour (see 3.5.2).

4.6.2 Distillation rate test. The no-load distillation rate shall be checked by measuring the condensed solvent as it returns to the storage tank to determine compliance with 3.5.1.1, 3.5.1.2, or 3.5.1.3, as applicable.

4.6.3 Vapor level control test. The recommended vapor level shall be automatically controlled and maintained to determine compliance with 3.6.4, 3.6.11, and 3.6.17.

4.6.4 Heat control test. The water inlet flow shall be turned off and when the solvent temperature has reached the preset temperature of the heat control, the electric heaters or steam supply shall be automatically shut off to determine compliance with 3.6.17.

4.6.5 Condenser assembly hydrostatic pressure test. With the condenser assembly (water jacket and coils) filled with water, and the outlet side (entering the open sight flow device) closed, a hydrostatic pressure of 40 psig shall be applied to the inlet side of the condenser assembly for a minimum of 30 minutes. Any evidence of leakage, damage to welds, seals, connections, or deformation of metal shall be cause for rejection (see 3.6.4).

4.6.6 Steam coils hydrostatic pressure test. The class 2 degreasers shall have the steam coils tested by closing the outlet side, filling with water, and subjecting them to a hydrostatic pressure of 75 psig for a minimum of 30 minutes. Any evidence of leakage or damage to the coils shall be cause for rejection (see 3.6.7).

4.6.7 Functional test. The degreasers shall remove the sludge and grease from 3000 pounds of metal parts, that have been stabilized at room temperature, in not more than one hour (see 3.5.2).

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4.6.8 Water temperature control test. While the degreasers are undergoing test, the automatic water control valve shall be tested to determine if it controls the water outlet of the condenser at its preset temperatures within ±5°F to determine compliance with 3.6.11.

4.6.9 Leakage. During any of the tests specified in 4.6.1 through 4.6.8, the degreasers shall show no evidence of leakage of any component.

4.6.10 Spray hose assembly and pump test. The spray hose assembly and spray pump shall be tested to determine compliance with 3.6.15 and 3.6.16.

* 4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Disassembly. The degreaser may be disassembled to the extent necessary to assure that all surfaces requiring a preservative are accessible for processing, to assure minimum cube for packing, and to permit reassembly without special skills or tools. Disassembly, when applicable, shall be as specified in MIL-STD-1186.

* 5.2 Preservation. Preservation shall be Level A or Industrial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Cleaning. Cleaning shall be accomplished in accordance with any process of MIL-P-116 that is non-injurious to the item.

5.2.1.2 Drying. Drying shall be accomplished in accordance with any applicable procedure of MIL-P-116.

5.2.1.3 Preservation application. Unpainted metal surfaces, and surfaces exposed by disassembly, shall be coated with type P-1 preservative in accordance with MIL-P-116. The entire water system shall be flushed with a solution consisting of 40 percent ethylene glycol conforming to O-A-548, type I, and 60 percent clean water. The interior surfaces of the spray pipe and nozzle shall be thoroughly drained.

* 5.2.1.4 Openings. Openings to the water and solvent inlets and outlets shall have means for closure provided or shall be sealed with tape conforming to type IV of PPP-T-60.

5.2.1.5 Intermediate packaging. The basic unit, with components and disassembled parts and tools, shall be packaged in accordance with method I of MIL-P-116.

5.2.1.5.1 Gages. All gages shall be covered with one-inch thick uncompressed cushioning material conforming to PPP-C-843, type II. The cushioning material shall be secured with the tape specified in 5.2.1.4. When gages are disassembled from the basic unit, they shall be packaged in accordance with method IA-15 of MIL-P-116.

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5.2.1.5.2 Spray hose assembly. Spray hose shall be packaged in accordance with Level A requirements of MIL-H-775.

5.2.1.5.3 Drive belt. The drive belt shall be either removed or released from tension. The face or grooves of ferrous metal pulleys shall be coated with rust-inhibiting primer conforming to TT-P-636. Belts removed from the degreaser shall be packaged in accordance with method III of MIL-P-116.

5.2.1.5.4 Electrical equipment. Electric motors shall be preserved and packaged in accordance with Level A requirements of MIL-E-16298 and the applicable subparagraph for assembled machines. Electric switch boxes shall be sealed and loose ends of the power cord shall be covered with tape conforming to PPP-T-60, type IV.

5.2.1.6 Consolidated package. Disassembled components, technical publications and packaged accessories shall be cushioned and packaged in one or more containers conforming to PPP-B-636, class weather-resistant, type CF. The consolidated packages and work basket shall be placed inside the cleaning tank and secured to the work supports with metal strapping conforming to QQ-S-781, type I. The metal cover shall be placed on top of the degreaser and secured with the above metal strapping or with strips of tape conforming to PPP-T-97.

* 5.2.2 Industrial. Industrial preservation and packaging shall be as specified in ASTM D3951-82.

5.3 Packing. Packing shall be Level A or Industrial, as specified (see 6.2).

5.3.1 Level A. The degreaser shall be packed in a crate conforming to MIL-C-52950, type V, style B, covered. Blocking, bracing, anchoring, and cushioning shall be in accordance with the container specification and MIL-STD-1186. Containers shall be closed in accordance with the container specification.

* 5.3.2 Industrial. Industrial packing shall be as specified in ASTM D3951-82.

* 5.4 Marking. Marking shall be Level A or Industrial, as specified (see 6.2).

* 5.4.1 Level A. Level A marking shall be as specified in MIL-STD-129.

* 5.4.2 Industrial. Industrial marking shall be as specified in ASTM D3951-82.

6. NOTES

6.1 Intended use. The vapor spray degreasers covered by this specification are intended for use in removing dirt, grease, oil, and sludge from parts.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Class and size required (see 1.2).
(c) First article when required (see 3.1).

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(d) Special safety devices when required (see 3.4.8 and 6.3).
(e) Freeboard height if different (see 3.6.18).
(f) Fungus control when required (see 3.6.20.3).
(g) Painting if different (see 3.7).
(h) Identification marking if different (see 3.8).
(i) First article inspection if different (see 4.3).
(j) Quality conformance inspection if different (see 4.4).
(k) Sampling when required (see 4.4.1).
(l) Level of preservation, packing, and marking required (see 5.2, 5.3, and 5.4).

6.3 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.4 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

6.6 Supersession data. This specification includes the requirements that were originally covered in Rock Island Arsenal Purchase Description T&EPD-28A, dated 15 November 1965.

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